

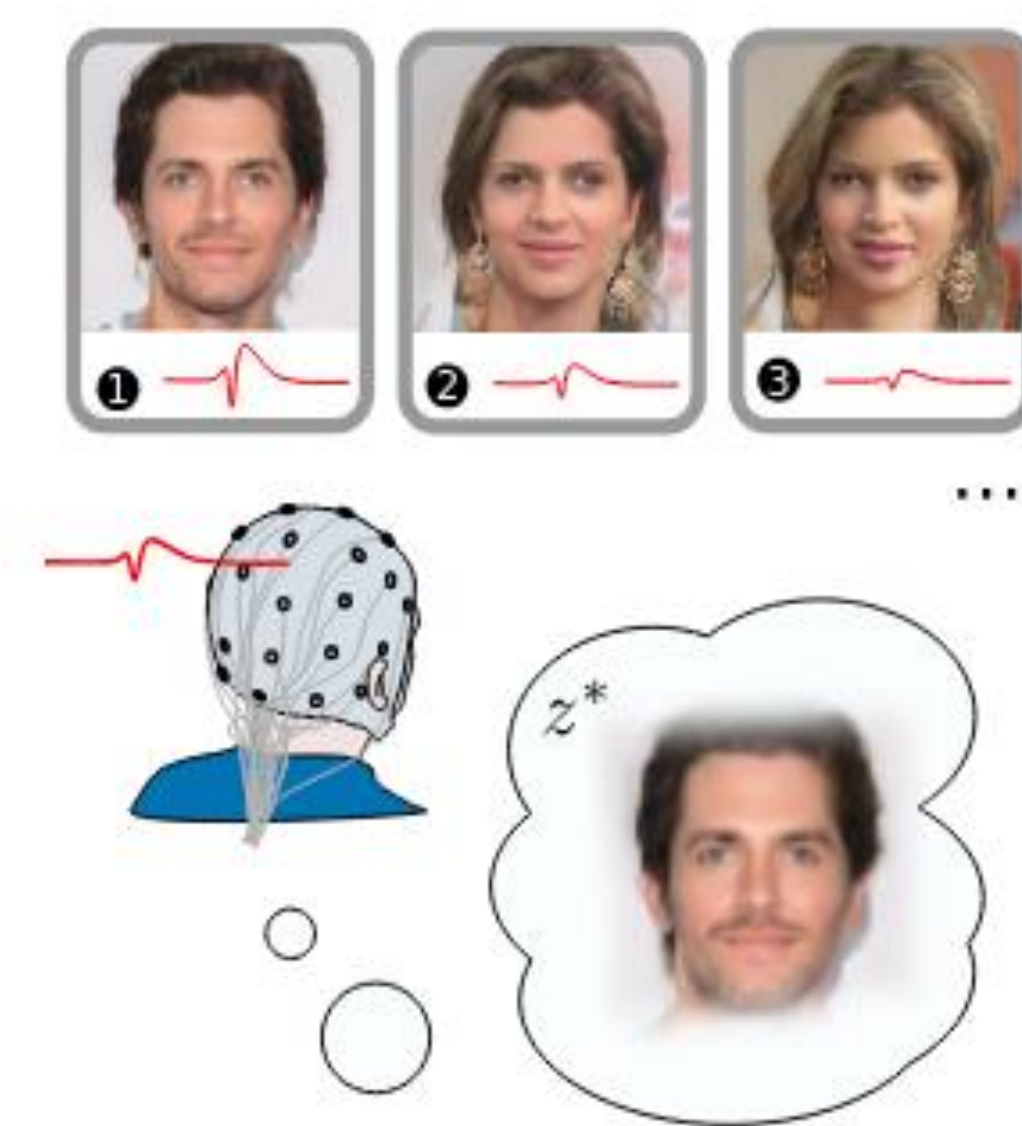
Self-Calibrating BCIs: Ranking and Recovery of Mental Targets Without Labels

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We recover the face image a participant holds in mind, using only unlabeled EEG responses

The self-calibration (SC) problem



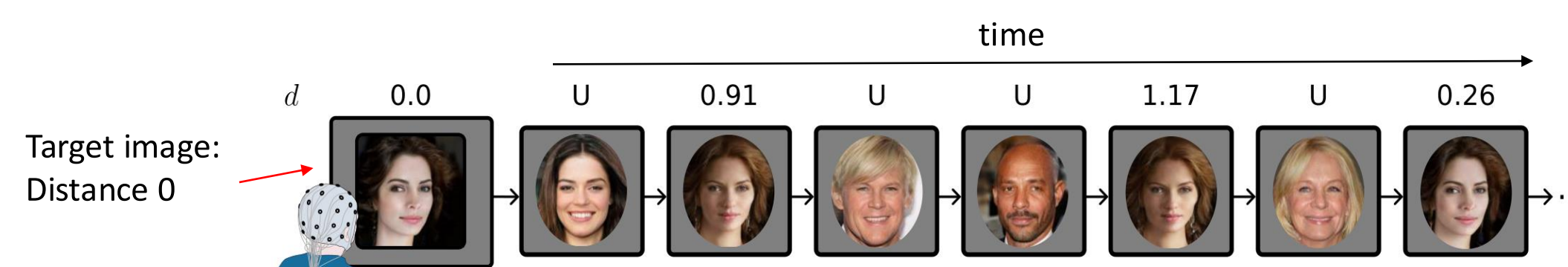
1) We give you unlabeled image and EEG pairs

2) You must recover the (face) image I had in mind

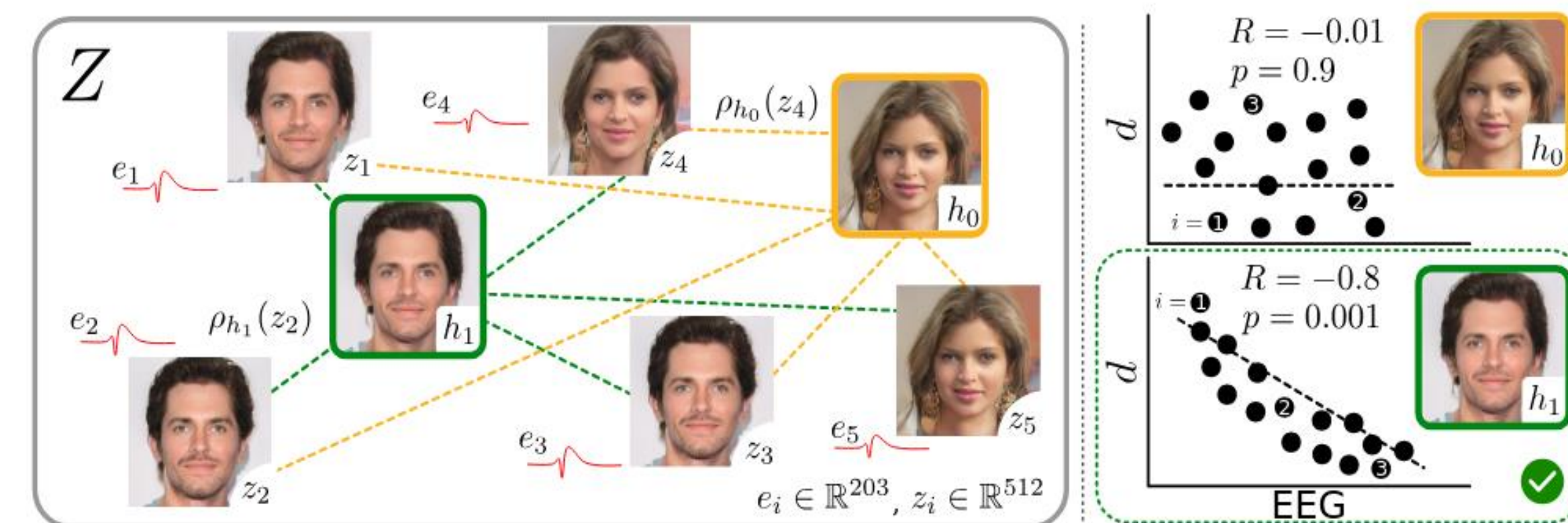
3) AND do it with:

"An estimator should have better prediction performance on a dataset 'labeled' from the 'point of view' of the image the user had in mind."

We contribute a new EEG dataset



We propose the continuous SC-BCI framework and solve it

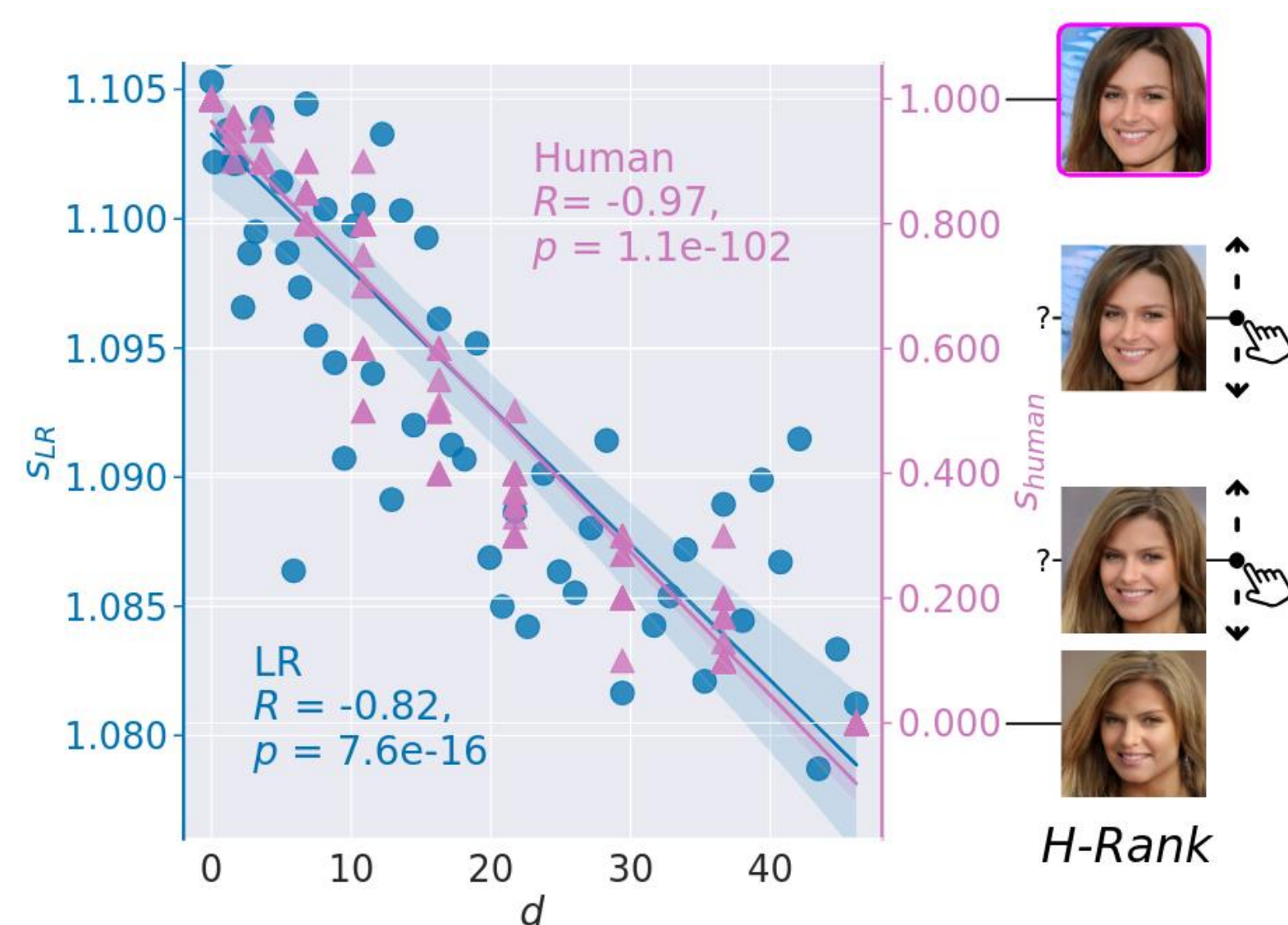


Limitations

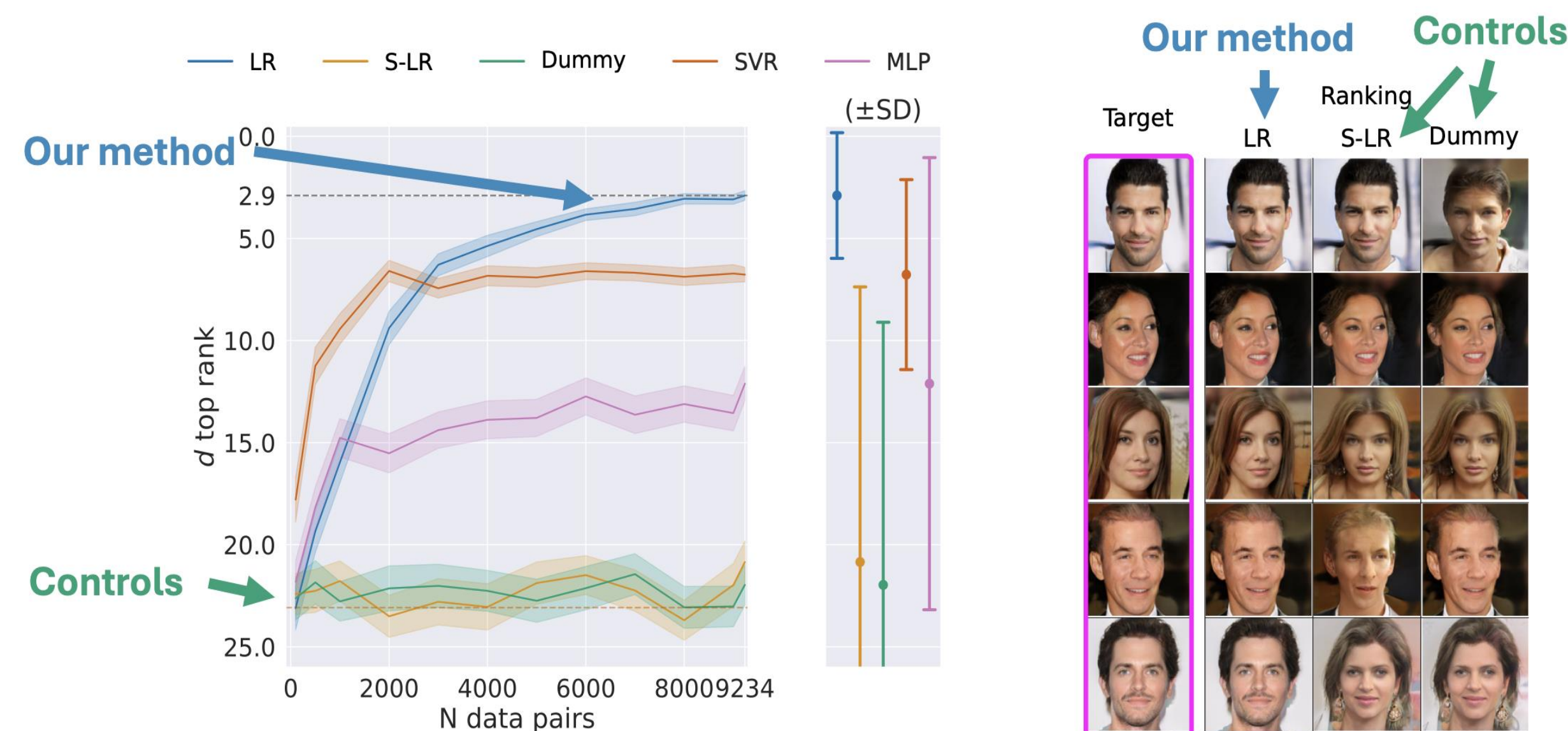
- This study is restricted to face stimuli from one pre-trained GAN.
- We relied on a predefined similarity function in the stimuli latent space.
- The optimization was run in a lower-dimensional latent subspace.
- Real-time performance remains untested.

Ethical considerations are critical as inferring intent without explicit consent raises privacy concerns.

Our scores correlate with human judgement



Top-1 retrieved faces are near the target



We can optimize and generate an *unknown* target

